

ITS-ShipFormer: An Informative Token Selection Former for SAR Ship Recognition

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Abstract—Ship automatic target recognition (ATR) in synthetic aperture radar (SAR) images plays a crucial role in maritime domain awareness. However, the sea clutter interference may make the model fail to focus on informative ship target regions for recognition. All existing SAR ship ATR models use the entire SAR ship image as input. But not all regions in SAR ship images contribute positively to recognition. However, the intraclass diversity and interclass similarity make the SAR ship ATR task more challenging. In this study, we propose a novel transformer-based architecture that addresses the two core challenges, named informative token selection former (ITS-ShipFormer). ITS-ShipFormer selects and guides the model's attention to the informative token of the ship target regions. The ITS-ShipFormer consists of the multihead dynamic local convolution (MHDLC) block in the early stages, transformer blocks equipped with a sea clutter suppression module (SCSM) in the latter stages, and a discriminative hybrid loss. SCSM automatically distinguishes the informative ship tokens and useless sea clutter tokens by two carefully designed strategies. In response to the other challenge, MHDLC is designed to enhance the feature extraction ability, and the hybrid discriminative loss is proposed to add constraints on the CLS token and the informative tokens simultaneously. The experimental results on benchmark datasets OpenSARShip and FUSAR-Ship jointly verify the effectiveness of our design. Different from previous works introducing the transformer structure in ITS-ShipFormer utilizes the overlooked innate and unique advantages of the transformer structure to address our challenge.

Index Terms—Attention map, effective and efficient, informative token, sea clutter interference, ship recognition, synthetic aperture radar (SAR), token score, transformer.

I. INTRODUCTION

IN RECENT years, the globalization of maritime trade significantly influenced the global economy, constituting approximately 80% of worldwide trade according to the United Nations Conference on Trade and Development [1], [2]. Ship recognition plays a crucial role in maritime surveillance, facilitating the identification of ship types to enhance the understanding of marine activities [3], [4], [5].

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Various ship recognition techniques are currently available, including visual monitoring, satellite remote sensing, and automatic identification system (AIS) monitoring [6]. While visual monitoring is easily deployable, it is weather-sensitive. The AIS monitoring system offers a wide monitoring range, low cost, and high real-time performance, but is limited to ships with AIS devices and susceptible to data tampering [7]. Synthetic aperture radar (SAR), utilizing radar technology for long-range imaging, is a method of remote sensing. SAR, unlike optical sensing, independently captures data regardless of lighting, weather, or cloud cover, offering distinct advantages in SAR ship automatic target recognition (ATR) [8], [9], [10].

There are SAR ship recognition methods based on traditional algorithms and convolutional neural networks (CNNs) algorithms. Traditional algorithms use manually extracted features (such as geometric features, scattering statistics features, and histogram of oriented gradient (HOG) features [11]) to recognize targets. After extracting the features, classifiers, such as support vector machine (SVM) [12], K -nearest neighbor [13], and random forest (RF) [14] are used for the subsequent recognition. However, these traditional algorithms have drawbacks, such as the need for artificial design of features and limited model scalability, making it challenging to meet the demands of intelligent SAR image analysis. Different from the traditional algorithms, CNNs utilize an end-to-end framework to achieve feature extraction and recognition, which has made great progress in the field of SAR ship ATR. For example, He et al. [15] designed a multitask learning CNN framework for better deep feature extraction and effective ship recognition in SAR images. Since each scale of features of the SAR ship is not equally effective for recognition, Shang et al. [16] and Wang et al. [17] enhanced the multiscale features from shallow to deep extracted by CNNs. Zeng et al. [18] and Xiong et al. [19] used dual-polarization features in the polarized channels (VV and VH) to further improve SAR ship recognition accuracy. Shang et al. [20] presented a CNN-based spherical space classifier to tackle the challenge that existing CNN classifiers are less powerful in extracting discriminative SAR ship features. Xie et al. [21] proposed a dual-branch deep network for ship classification of dual-polarized SAR images. But all these deep learning models may concentrate on useless or even harmful regions to conduct recognition [22], [23]. In recent years, some explorations [6], [24], [25], [26] try to incorporate traditional mature hand-crafted features to aid the model in making more effective and interpretable ship recognition decisions. For instance,

Zhang et al. [24] proposed HOG-ShipCLSNet, which utilizes features extracted by the HOG algorithm and CNN to realize preferable SAR ship recognition. Zheng et al. [25] introduced the scattering features into the deep network and characterized the features of ship targets more comprehensively. However, these explorations have a fatal drawback, needing the SAR ship images with a clear background to guarantee the effectiveness of the extracted hand-crafted features.

Recently, vision transformer (ViT) [27] has achieved advanced performance in image recognition. By utilizing a multihead self-attention (MSA) mechanism, the transformers can establish long-range relationships between each image token, showing a more powerful representation capability compared to CNNs. SAR images are different from optical images due to their specific imaging process. Each scattering point of the SAR image is related to a larger range of the actual scene. Hence, transformer structures are inherently well-suited as feature extraction models for SAR images [22], given the requirement for constructing long-distance information in SAR image feature extraction models. However, to the best of the authors' knowledge, only limited works introduced transformer structures into the SAR ship ATR task. The works exploring transformers on SAR images have not been fully started [28].

Based on the above analysis, it can be concluded that SAR ship ATR has achieved great breakthroughs. However, achieving further SAR ship accuracy improvements still faces certain challenging issues, which are shown in Fig. 1.

- 1) *Sea Clutter Interference*: Excluding the interference of sea clutter and directing the model's attention toward ship target regions poses a challenging yet imperative problem that necessitates resolution, but no one has explored it. The recognition of maritime targets (i.e., ships) inevitably faces interference from sea clutter. Due to the influence of environmental factors, ship wakes, radar equipment parameters, and various incidence angles¹ for SAR radar imaging of the sea surface, sea clutter exhibits complex randomness and variability, adding to the complexity of maritime SAR images.
- 2) *Intraclass Diversity and Interclass Similarity*: Due to the complex imaging conditions and diverse viewing angles between ship targets and SAR sensors, ships in SAR images exhibit varying aspects in the azimuth direction and distinct scattering characteristics. Furthermore, ships within the same class may display structural differences, while those from different classes may share similar sizes and topside structures. Due to the impacts of these factors, intraclass diversity and interclass similarity are present.

To address the issue of sea clutter interference, existing algorithms can generally be categorized into two groups: 1) designing SAR speckle filters [29], [30], [31] grounded in mathematical theory to enhance noise suppression and

2) training deep CNNs [32], [33] to smooth speckles while preserving image details. Although these advanced methods can effectively mitigate such effects, they suffer from a critical limitation—being difficult to integrate seamlessly into end-to-end recognition networks. Even when models extract features from filtered SAR ship images, they might still focus on noninformative or even misleading regions during recognition. As for the intraclass diversity and interclass similarity issue, most existing advanced methods try to extract more discriminative features and design an impressive loss function. As for the challenge of intraclass diversity and interclass similarity, most existing advanced approaches focus on extracting more discriminative features [34], [35], and designing sophisticated loss functions [22]. Inspired by these studies, this work also addresses the problem from both perspectives.

In this article, we propose a task-specific SAR ship ATR transformer architecture called informative token selection former (ITS-ShipFormer) to deal with the challenges in SAR ship ATR while maintaining our model's effectiveness and efficiency. ITS-ShipFormer is built up with four stages. Different from previous works introducing transformer models into the SAR ATR community for establishing long-range feature correlations, the proposed ITS-ShipFormer utilizes the overlooked innate and unique advantages of the transformer structure, i.e., without any additional annotations, the transformer structure encoded with position information can directly capture the importance of each image token regarding the class information, thus selecting and localizing the informative tokens for discriminative feature extraction.

In ITS-ShipFormer, to deal with the challenge of sea clutter interference and direct the model's attention toward target regions, a self-motivated sea clutter suppression module (SCSM) is proposed and inserted into the transformer blocks to guide the model in distinguishing between the useless tokens, consisting of worthless or harmful sea clutter regions, and the informative tokens, consisting of discriminative SAR ship target regions. The SCSM consists of two components: informative token selection and double flow token updating. The informative token selection precisely distinguishes the two token types, decreasing the influence of sea clutter interference. The double flow token updating strategically focuses computations on informative tokens through differentiated updates, ensuring both the effectiveness and efficiency of ITS-ShipFormer.

To address intraclass diversity and interclass similarity, we propose a multihead dynamic local convolution (MHDLC) block to enhance the feature extraction ability. Rather than substituting MSA with standard convolution, the MHDLC block replaces MSA with multihead dynamic convolution, emulating MSA to extract more discriminative SAR ship features. The multihead dynamic convolution can dynamically allocate weights over different spatial positions so as to emphasize the most informative features and alleviate the redundancy of kernels by dividing the channel dimension into groups and sharing kernel weights within each group, also ensuring the effectiveness and efficiency simultaneously of ITS-ShipFormer.

¹As shown in the last SAR image of the first row in Fig. 1, when SAR radar images the sea surface with a small incidence angle, it may cause strong interference from the sea background.

Finally, the ITS-ShipFormer is updated using a discriminative hybrid loss to optimize our model with increased inner-class compactness and interclass separability. The innovations of our method are summarized as follows.

- 1) We introduce a novel transformer model, ITS-ShipFormer, for two challenging issues in SAR ship ATR. To the best of the authors' knowledge, this is the first attempt to address the challenge of sea clutter interference while learning to focus on ship regions using an end-to-end model with only image-level supervision. However, ITS-ShipFormer provides a completely new perspective to comprehend and utilize the distinctive and innate advantage of the transformer structure for SAR image understanding.
- 2) In response to the sea clutter interference and the model's failure to concentrate on target region issues, SCSM is proposed and inserted into the transformer blocks to guide the model's attention to the SAR ship regions by distinguishing the informative tokens and useless tokens. SCSM employs two effective strategies to ensure the accuracy of informative tokens and utilizes a double flow token updating mechanism to prioritize computational resources toward significant feature regions.
- 3) In response to the intraclass diversity and interclass similarity issue, two key enhancements are proposed. First, we design an MHDLC block that utilizes a multihead dynamic convolution to emulate the MSA mechanism and extract features dynamically, enabling the enhancement of the feature extraction ability. Second, a discriminative hybrid loss is proposed, which adds constraints on the CLS token of the global image and the selected informative tokens with local richness ship features simultaneously, forcing the model to extract features with increased inner-class compactness and interclass separability.

The ITS-ShipFormer achieved the SOTA performance in the recognition of the benchmark OpenSARShip [36] and FUSAR-Ship [37] datasets. Extensive experiments and visualizations are performed to validate the effectiveness of our methods.

The rest of this article is arranged as follows. The details of the proposed ITS-ShipFormer are presented in Section II. Section III validates the effectiveness of the proposed method through extensive experiments and visualizations. Finally, conclusions are presented in Section IV.

II. METHODOLOGY

In this section, the proposed task-specific ITS-ShipFormer model is described in detail to tackle the challenges in SAR ship ATR, and the MHDLC block, SCSM, and discriminative hybrid loss are presented.

A. Framework of ITS-ShipFormer

Achieving satisfactory SAR ship ATR performance hinges on addressing two key challenges: sea clutter interference,

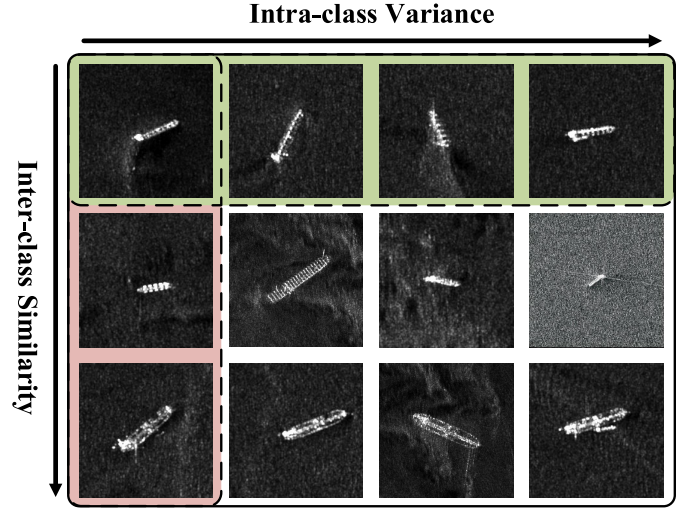


Fig. 1. Key challenges for ship recognition in SAR images. Each row corresponds to the same class for SAR ship ATR.

which makes the ATR models fail to capture informative ship regions for recognition, and intraclass diversity and interclass similarity, which make the SAR ship ATR more challenging. Hence, ITS-ShipFormer is designed to select and locate informative ship target regions, thereby eliminating sea clutter interference, while also focusing on optimal feature extraction and constructing a discriminative loss function to address intraclass diversity and interclass similarity issues. Through this process, the model adaptively forces it to capture informative ship target regions and extract optimal features, increasing inner-class compactness and interclass separability, ultimately leading to improved SAR ship recognition performance.

As shown in Fig. 2, our proposed ITS-ShipFormer consists of MHDLC blocks in the first two stages and transformer blocks equipped with SCSM in the last two stages, along with the discriminative hybrid loss, adding constraints on both CLS token and informative tokens to optimize our model. The ITS-ShipFormer pipeline is presented below.

The input SAR ship images are first split into nonoverlapping patches with a patch size of 4×4 and then input into the MHDLC blocks to utilize multihead dynamic convolution to extract features effectively. Next, in the latter two stages of ITS-ShipFormer, the transformer blocks equipped with SCSM can identify the informative tokens and useless tokens. A double flow token updating manner is employed in SCSM to update the informative tokens and useless tokens differently, concentrating the computations on informative tokens and decreasing the influence of the sea clutter interference. Finally, two different loss functions are employed on the CLS token and informative tokens to collaboratively update our model, helping to enhance the inner-class compactness and interclass separability.

We can note that the precision of the informative tokens is paramount for our model's performance in SAR ship ATR, significantly influencing the effectiveness of both SCSM and the loss function. A mixed attention scoring strategy and a global attention evolution strategy are employed in SCSM to

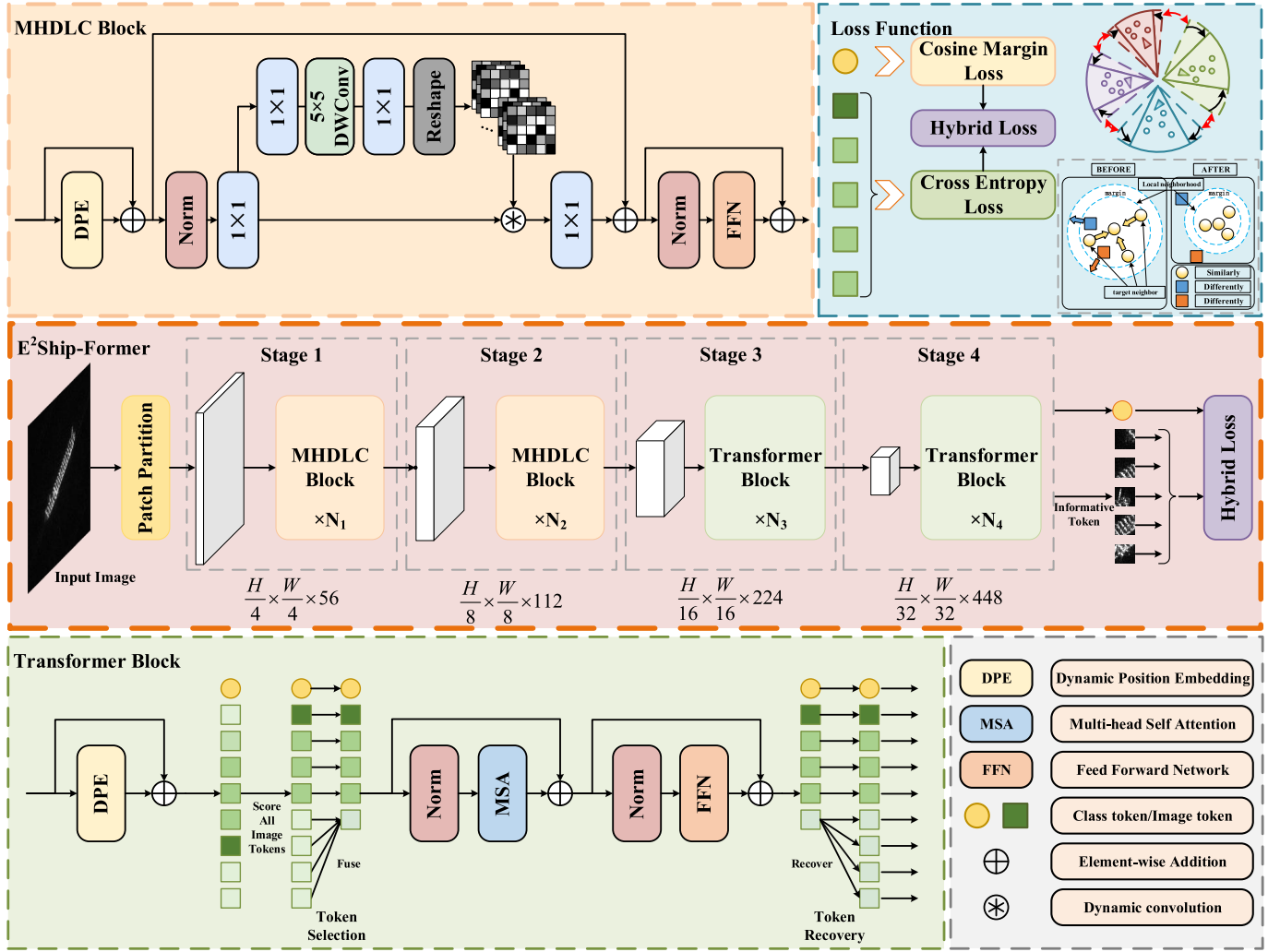


Fig. 2. Architecture of ITS-ShipFormer, which comprises four successive stages for effective and efficient SAR ship feature extraction and a hybrid loss to enhance the feature's inner-class compactness and interclass separability. The pipeline of ITS-ShipFormer is presented in the middle. The first two stages of ITS-ShipFormer are comprised of MHDLC blocks, whose detailed structure can be found at the top-left corner. Transformer blocks equipped with SCSSM are employed in the last two stages, whose detailed information is located in the bottom-left corner. Finally, how the loss function works is illustrated in the top-right corner. The cosine margin loss is applied on the CLS token to optimize the extracted global information, and the CE loss is utilized on all informative tokens to comprehensively exploit the richness of local information.

guarantee the precision of the informative tokens. The mixed attention scoring strategy is first utilized to calculate the attention scores of each image token. Then, for better capability of capturing the underlying information among tokens in different layers, the global attention evolution strategy is employed to further guarantee the precision of the informative tokens by evolving our mixed attention scoring strategy across layers. Through these strategies and the innovations aforementioned, ITS-ShipFormer can improve the SAR ship ATR performance with sea clutter interference and intraclass diversity and interclass similarity challenges in a targeted manner.

B. MHDLC Block

SAR images exhibit intraclass diversity (e.g., varying ship orientations) and interclass similarity (e.g., similar ship structures), making feature extraction challenging. Recent studies replace the MSA mechanism with standard convolution in the early stages of the transformer structure, achieving

satisfactory performance and improving efficiency. Nevertheless, using standard convolution undeniably entails a compromise in feature extraction prowess.² This compromise hinders the extraction of optimal features necessary to address the complexities of intraclass diversity and interclass similarity in SAR ship ATR. The MHDLC block enhances the model's ability to extract discriminative features by replacing standard MSA with a multihead dynamic convolution³ mechanism tailored for SAR images. Unlike traditional convolutions with fixed kernels, MHDLC dynamically adjusts kernel weights based on input token features, emphasizing ship-specific details.

²Standard convolution reuses the same kernels among different positions in the spatial domain, which deprives the ability to adapt to diverse features with respect to different positions.

³Inspired by MSA, we divide the channel dimension of the dynamic convolution C into G groups to learn separate dynamic weights in parallel.

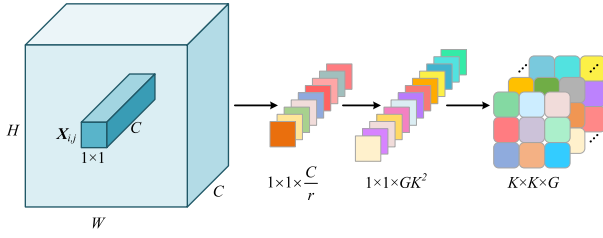


Fig. 3. Schematic of a multihead dynamic convolution tailored for pixel $X_{i,j} \in \mathbb{R}^C$. G counts the number of groups where each group shares the same dynamic kernel across the channel dimension.

The pipeline of the proposed MHDLC can be found at the top-right corner in Fig. 2. MHDLC first estimates the multihead dynamic convolution to explore local features dynamically. Then, the estimated multihead dynamic convolution kernel is used as the dynamic local attention and applied to input features for better feature aggregation, thereby enhancing the discriminative features. Finally, similar to a transformer, a feedforward network is employed to improve feature representation [27].

Given a feature map $X \in \mathbb{R}^{H \times W \times C}$ generated by a layer norm followed by a 1×1 convolution, the multihead dynamic convolution kernel $\mathcal{H} \in \mathbb{R}^{H \times W \times K \times K \times G}$ is devised. G counts the number of groups where each group shares the same dynamic kernel across the channel dimension. Specially, $\mathcal{H}_{i,j} \in \mathbb{R}^{K \times K \times G}$ denotes a dynamic kernel with size $K \times K$, which is especially tailored for pixel $X_{i,j} \in \mathbb{R}^C$ located at the corresponding coordinate (i, j) in $X \in \mathbb{R}^{H \times W \times C}$. The schematic of a dynamic kernel $\mathcal{H}_{i,j}$ is shown in Fig. 3, which generated achieved by

$$X'_{i,j} = \text{DWConv}_{5 \times 5}(\text{Conv}_{1 \times 1}(X_{i,j})), X'_{i,j} \in \mathbb{R}^{C/r} \quad (1)$$

$$X''_{i,j} = \text{Conv}_{1 \times 1}(X'_{i,j}), X''_{i,j} \in \mathbb{R}^{GK^2} \quad (2)$$

$$\mathcal{H}_{i,j} = \mathcal{R}(X''_{i,j}), \mathcal{H}_{i,j} \in \mathbb{R}^{K \times K \times G} \quad (3)$$

where r denotes the reduction ratio for efficient processing; $\text{DWConv}_{5 \times 5}$ denotes a depthwise convolution with filter size of 5×5 pixels; $\text{Conv}_{1 \times 1}$ denotes a convolution with filter size of 1×1 pixel; $\mathcal{R}$ denotes the reshape operation. The depthwise convolution ensures the dynamic kernel can better model the local information.

With the generated pixelwise dynamic kernel $\mathcal{H}_{i,j}$, the corresponding output feature $Y_{i,j}$ is obtained by

$$Y_{i,j} = \mathcal{H}_{i,j} \otimes X \quad (4)$$

where $\otimes$ denotes the dynamic convolution operation with a weight-sharing mechanism in each group of channels [38]. The details of (6) is defined as

$$Y_{i,j,k} = \sum_{(u,v) \in \Delta_K} \mathcal{H}_{i,j,u+[K/2],v+[K/2],[kG/C]} X_{i+u,j+v,k} \quad (5)$$

where $k \in [1, 2, \dots, C]$, $\Delta_K \in \mathbb{Z}^2$ refers to the set of offsets in the neighborhood considering dynamic convolution on the center pixel, written as

$$\Delta_K = [-\lfloor K/2 \rfloor, \dots, \lfloor K/2 \rfloor] \times [-\lfloor K/2 \rfloor, \dots, \lfloor K/2 \rfloor]. \quad (6)$$

Besides effectiveness, multihead dynamic convolution can also extract features in a more efficient way. In the channel

domain, standard convolution uses different kernels among different channels, causing stand-out interchannel redundancy [38]. On the contrary, inspired by MSA, the multihead dynamic convolution mitigates the computation redundancy by dividing the channel dimension into groups and sharing kernel weights within each group. Through this manner, the MHDLC block ensures the effectiveness and efficiency of ITS-ShipFormer simultaneously.

C. Self-Motivated SCSM

Sea clutter often obscures ship targets in SAR images, causing models to focus on irrelevant regions. The SCSM addresses this by identifying and prioritizing informative tokens (those representing ship targets) while suppressing useless tokens (those dominated by sea clutter). Think of SCSM as a filter that sifts through a noisy image to highlight only the regions critical for recognition. Fig. 4 provides a detailed flowchart of SCSM's workflow.

The SCSM comprises two components: 1) informative token selection and 2) double flow token updating. For informative token selection, the mixed attention scoring strategy and global attention evolution strategy are employed in SCSM to guarantee the precision of informative tokens. However, a double flow token updating strategy is applied to two token types. The expressive slow path performs feature extraction carefully on informative tokens. The fast path utilized on useless tokens reduces the computational complexity and decreases the influence of the sea clutter interference. Through this manner, the effectiveness and efficiency of ITS-ShipFormer are ensured simultaneously.

1) Informative Token Selection: The attention map is the “golden heart” of the transformer block and serves as the most intuitive approach to interpreting the model’s decision and gaining insights about the propagation of information among tokens, serving as the foundation of informative token selection. The mixed attention scoring strategy and global attention evolution strategy are all based on utilizing the “golden heart.” Here, the mixed attention scoring strategy is first outlined for calculating the attention scores of each image token. Subsequently, to enhance the model’s capability to capture underlying information among tokens in different layers, a global attention evolution strategy is proposed. This strategy further ensures the precision of informative tokens by evolving our mixed attention scoring strategy across layers.

As shown in the right part of Fig. 4, the mixed attention scoring strategy is composed of two parts: CLS-to-tokens and tokens-to-tokens. In CLS-to-tokens, only the CLS token can “vote” to decide which tokens are informative tokens, while in tokens-to-tokens, each image token can “vote.” We first explain the CLS-to-tokens in detail.

The attentive scores between the CLS token and image tokens are described in (2). The distinguished informative tokens have high similarities (i.e., attention scores) with the CLS token, while the useless tokens have low similarities [39]. We term the strategy of only utilizing the CLS token for identifying informative tokens as the CLS-to-tokens strategy. However, each token encapsulates abundant information from its corresponding local image patch. Relying solely on the

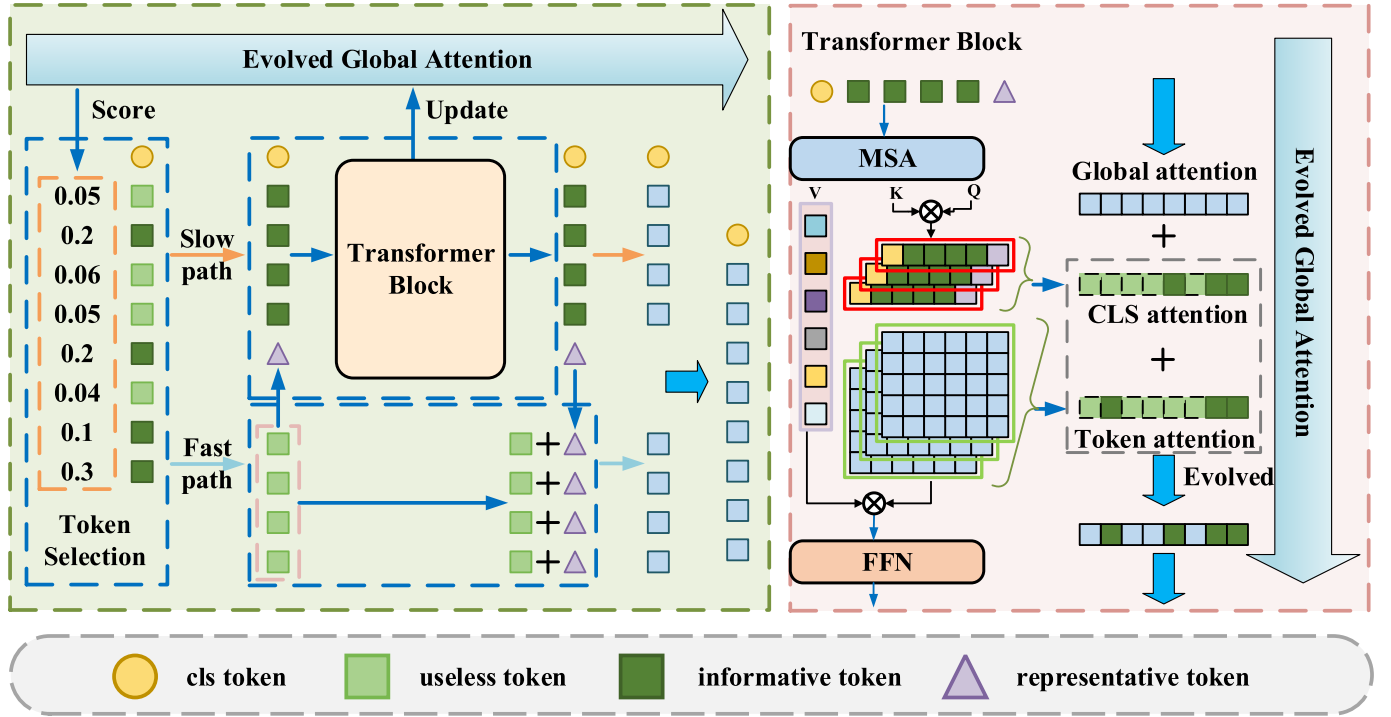


Fig. 4. Overall framework of self-motivated SCSM. The SCSM module is inserted into the vanilla transformer block to adaptively distinguish the informative tokens and useless tokens. As shown in the left part, by applying a lightweight fast path to useless tokens and an expressive slow path to informative tokens, the computations can be reduced, and the features of informative tokens are effectively extracted. In addition, the precision of the informative tokens is vital to our model. As shown in the right part, to guarantee the precision, a mixed attention scoring strategy and a global attention evolution strategy are employed.

CLS token for recognition is insufficient to fully exploit the wealth of information available. Thus, another tokens-to-tokens strategy is introduced as supplementary. We combine these two strategies to distinguish the informative tokens and useless tokens, denoting the attentive scores calculated by the CLS-to-tokens strategy as A_{cls} , the attentive scores calculated by the tokens-to-tokens strategy as A_{t2t} , and the fused attentive scores as A_{mix} . Mathematically, the attentive scores are formulated as

$$A_{cls} = \text{Softmax} \left(\frac{q_{cls} \cdot K^T}{\sqrt{C}} \right) \quad (7)$$

$$A_{t2t} = \frac{1}{N} \sum_{i=1}^N \text{Softmax} \left(\frac{q_{token_i} \cdot K^T}{\sqrt{C}} \right) \quad (8)$$

$$A_{mix} = A_{cls} + \lambda A_{t2t} \quad (9)$$

where q_{cls} is the query vector of the CLS token, q_{token_i} is the query vector of a image token, N is the number of image tokens, and λ is the mixing weight hyperparameter. By calculating the attentive scores A_{mix} , we select k tokens whose scores in A_{mix} are among the top k as the informative tokens. The remaining $N - k$ tokens are recognized as useless tokens that contain sea clutter.

For better capability of capturing the underlying information among tokens in different layers, we propose a global attention that augments the mix attention A_{mix} by evolving it across layers. Specifically, a residual connection between A_{mix} of different layers is designed to facilitate the attention

information flow. Mathematically

$$A_{global}^l = \alpha \cdot A_{global}^{l-1} + (1 - \alpha) \cdot A_{mix}^l \quad (10)$$

where A_{global}^l is the global attention in the l th layer, and A_{mix}^l is the mix attention in the l th layer. We use A_{global}^l for the token selection in the $(l + 1)$ th layer for stability and efficiency. For each layer with token selection, only the global attention scores of the selected informative tokens are updated.

2) *Double Flow Token Updating*: As shown in the left part of Fig. 4, once the informative tokens and the useless tokens are determined, we utilize a double flow token updating approach to update the informative tokens and useless tokens differently, concentrating the computations on informative tokens and guaranteeing the efficiency of our model.

In the slow updating path, informative tokens are carefully updated via MSA and FFN, while useless tokens are summarized and updated via a representative token. The slow token updating strategy is introduced mathematically as follows.

Given the selecting ratio ρ , the number of informative tokens in the l th layer are

$$k^l = \lfloor \rho N \rfloor. \quad (11)$$

Formally, for the l th layer, let

$$\mathcal{I}^l = \text{argsort} (A_{global}^l) \quad (12)$$

where the argsort denotes calculating the indices sorted by the A_{global}^l with a descending order.

We then select the tokens of which the indices lie in $\mathcal{I}_{1:k}^l$ as informative tokens $x_{\text{info}} \in \mathbb{R}^{k \times C}$, and which lie in $\mathcal{I}_{k+1:N}^l$ as useless tokens, $x_{\text{ul}} \in \mathbb{R}^{(N-k) \times C}$

$$\begin{aligned}\mathcal{I}_{\text{info}}^l &= \mathcal{I}_{1:k}^l \\ \mathcal{I}_{\text{ul}}^l &= \mathcal{I}_{k+1:N}^l.\end{aligned}\quad (13)$$

Then, the useless tokens x_{ph} are aggregated into a representative token $x_{\text{rep}} \in \mathbb{R}^{1 \times C}$, as follows:

$$x_{\text{rep}} = \phi_{\text{agg}}(x_{\text{ul}}) = \sum_i \frac{A_{\text{global}}^l(\mathcal{I}_{uli}^l)}{\sum_j A_{\text{global}}^l(\mathcal{I}_{ulj}^l)} \cdot x_{uli} \quad (14)$$

where $\phi_{\text{agg}} : \mathbb{R}^{(N-k) \times C} \rightarrow \mathbb{R}^{1 \times C}$ denotes an aggregating function. Here we use a weighted sum based on the corresponding global attention score A_{global}^l .

Then, the CLS token, the informative tokens x_{info} and the representative token x_{rep} are fed into MSA and FFN modules, and their residuals are recorded as $x_{\text{inf}}^{(*)}$ and $x_{\text{rep}}^{(*)}$ for skip connections, which can be denoted by

$$\begin{aligned}x &= [x_{\text{cls}}, x_{\text{info}}, x_{\text{rep}}] \\ x_{\text{cls}}^{(1)}, x_{\text{info}}^{(1)}, x_{\text{rep}}^{(1)} &= \text{MSA}(x) \\ x &\leftarrow x + [x_{\text{cls}}^{(1)}, x_{\text{info}}^{(1)}, x_{\text{rep}}^{(1)}] \\ x_{\text{cls}}^{(2)}, x_{\text{info}}^{(2)}, x_{\text{rep}}^{(2)} &= \text{FFN}(x) \\ x &\leftarrow x + [x_{\text{cls}}^{(2)}, x_{\text{info}}^{(2)}, x_{\text{rep}}^{(2)}].\end{aligned}\quad (15)$$

Thus, the CLS token x_{cls} , the informative tokens x_{info} , and the representative token x_{rep} are updated in a slow and elaborate way.

After finish the slow updating, the useless tokens x_{ul} are updated in a fast way by the residuals of x_{rep}

$$x_{\text{ul}} \leftarrow x_{\text{ul}} + \phi_{\text{exp}}(x_{\text{rep}}^{(1)}) + \phi_{\text{exp}}(x_{\text{rep}}^{(2)}) \quad (16)$$

where $\phi_{\text{exp}} : \mathbb{R}^{1 \times C} \rightarrow \mathbb{R}^{(N-k) \times C}$ denotes an expanding function, such as a simple copy in our method.

D. Discriminative Hybrid Loss

To further improve recognition accuracy, ITS-ShipFormer uses a discriminative hybrid loss that optimizes both global and local features. This loss combines two components as follows.

- 1) *CLS Token Loss*: A cross-entropy (CE) loss is applied to the CLS token, which aggregates global image information, ensuring overall classification accuracy.
- 2) *Informative Token Loss*: A cosine-margin loss is applied to the selected informative tokens, encouraging compact feature representations within the same ship class and separable features between different classes [40].

This hybrid loss enhances inner-class compactness and interclass separability, making the model robust to the complex variations in SAR ship images.

Given an input feature vector x_i with its corresponding label y_i , the basic CE loss can be formulated as

$$L(x_i, y_i) = -\log \frac{e^{f_{y_i}}}{\sum_{j=1}^C e^{f_j}} \quad (17)$$

where C is the number of classes, and f_j is the output vector of the last fully connected layer before CE with weight vector W_j

$$f_j = W_j^T x = \|W_j\| \|x\| \cos \theta_j. \quad (18)$$

The cosine margin loss requires the $\|W_1\| \|x\| (\cos \theta_1 - m) > \|W_2\| \|x\| \cos \theta_2$ and $\|W_2\| \|x\| (\cos \theta_2 - m) > \|W_1\| \|x\| \cos \theta_1$. Here, θ is the angle between W and x , and m is a positive integer control the magnitude of the cosine margin. When m increases, the recognition margin becomes larger, and the learning objective becomes harder. We normalize $\|W_j\| \|x\|$ to a fixed scale s . Then, different from CE loss, the equation of cosine margin loss is present as

$$L_{\text{cm}}(x_i, y_i) = -\log \frac{e^{s(\cos(\theta_{y_i, i}) - m)}}{e^{s(\cos(\theta_{y_i, i}) - m)} + \sum_{j \neq y_i} e^{s \cos(\theta_{j, i})}}. \quad (19)$$

The cosine margin loss applied to the CLS token is denoted as $L_{\text{cm}}(X_{\text{CLS}}, y_i)$.

For local informative tokens, CE loss $L(\cdot, \cdot)$ is employed between each informative token X_{ij} and corresponding label y_i . Given the informative tokens $X_{i1}, \dots, X_{ik}$, the loss applied on the informative tokens is formulated as

$$L_{\text{ce}}(X_i, y_i) = \frac{1}{k} \sum_{j=1}^k L(X_{ij}, y_i). \quad (20)$$

Finally, the discriminative hybrid loss of ITS-ShipFormer can be achieved by combining L_{cm} and L_{ce}

$$L_{\text{E2Ship-Former}} = L_{\text{cm}}(X_{\text{CLS}}, y_i) + \beta L_{\text{ce}}(X_i, y_i) \quad (21)$$

where β is a hyperparameter to balance the two losses.

III. EXPERIMENTS AND RESULTS

In this section, the effectiveness of the proposed method is evaluated on the benchmark OpenSARShip and FUSAR-Ship datasets. The datasets, experimental settings, and evaluation indices are introduced in detail. The experimental results are demonstrated for OpenSARShip under a three-category recognition task and FUSAR-Ship under a seven-category recognition task, respectively. In addition, a series of ablation studies is conducted. Visualizations of the informative tokens selected by our model and heatmaps of the SAR ship images with and without the proposed method are then presented to validate the soundness of the method. Finally, we perform a comprehensive comparison between the proposed method and some SOTA methods. We also compare the implementation costs of the SOTA methods with our model to demonstrate its efficiency. All the experiments are implemented on a personal computer (PC) with NVIDIA GeForce RTX 2060 VENTUS (12 GB) GPU and 24 GB RAM. Software development is conducted under the Python language environment using an open-source machine learning library PyTorch. CUDA10.1 is used for training and inference acceleration.

A. Dataset Description

To evaluate the effectiveness of ITS-ShipFormer, we conducted a comprehensive experimental analysis on two benchmark SAR ship datasets, like other researchers [16],

TABLE I
DISTRIBUTION OF THE SAR SHIP DATASETS

Dataset	Category	Train	Test	All
OpenSARShip	Bulk carrier	338	328	666
	Container ship	338	808	1146
	Tanker	338	146	484
FUSAR-Ship	Container ship	1219	523	1742
	General cargo	1205	517	1722
	Fishing	1101	473	1574
	Tanker	1215	521	1763
	Bulk carrier	1150	494	1644
	Other cargo	1214	520	1734
	Others	1211	521	1732

[25]. The datasets' distribution ratio aligns with the popular SOTA methods [20], [24], [25]. Table I presents the dataset distribution details.

1) *OpenSARShip*: The OpenSARShip dataset comprises dual-polarization SAR imagery captured by the European Space Agency's Sentinel 1 satellite, encompassing both VH and VV polarization channels. Leveraging the coordinate and category information provided by Huang et al. [36], and following the experimental protocols established in previous studies, we extract three primary ship types. The dataset adopted in this experiment is the ground range detected (GRD) product under Sentinel-1 IW mode with 20×22 m resolution and 10.0×10.0 m pixel size in azimuth and range directions.

2) *FUSAR-Ship*: The high-resolution FUSAR-Ship dataset is derived from the Gaofen-3 (GF-3) satellite, marking China's inaugural civil C-band fully polarimetric spaceborne SAR. GF-3 SAR images exhibit an azimuth resolution of 1.124 m and a slant range resolution ranging from 1.700 to 1.754 m. This dataset is meticulously curated through an automated SAR-AIS matchup procedure encompassing more than 100 GF-3 scenes, featuring over 5000 ship image chips with accompanying AIS information.

B. Parameter Settings and Evaluation Metrics

1) *Parameter Settings*: These experiments are trained under the same parameters. The size of the input images, both in OpenSARShip and FUSAR-Ship are unified to 224×224 . Using stochastic gradient descent (SGD) optimizer with the weight decay parameter 0.0005 and the momentum parameter 0.9, the proposed network is trained for 100 epochs. The batch size is set to an empirical value of 16 due to the limited GPU memory. To mitigate the negative effects of vanishing training gradients, we set the learning rate to a relatively small value of 0.0005. For ITS-ShipFormer, the block numbers of each stage N_1 , N_2 , N_3 , and N_4 are set as 2, 5, 8, and 2. In the MHDLC block, the value of the size of the dynamic kernel K is set as 5, and the number of groups G is set as 16. In SCSM, the mixing weight λ and global weight α are set as 0.5. The selection ratio is set to 0.1. In the hybrid loss function,

the values of scale s , margin m , and β are set as 5, 0.1, and 1, respectively.

2) *Evaluation Metrics*: Accuracy (Acc) is used as the core evaluation criterion to measure recognition performance. Acc is the ratio of the number of correctly recognized samples to the total number of samples, which is defined by

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (22)$$

where TP denotes the true positives, TN denotes the true negatives, FP denotes the false positives, and FN denotes the false negatives. Furthermore, three other performance metrics are employed for more result verifications in the experiments, including as follows: 1) Precision; 2) Recall; and 3) $F1$.

C. Recognition Results

1) *OpenSARShip*: Table II presents the recognition performance of the proposed method on the OpenSARShip under three categories for each ship class in confusion matrix form. It is observed that the proposed ITS-ShipFormer achieves an accuracy of 84.09%. The diagonal values are significantly higher than others in the same line from Table II, which indicates that most ships can be recognized correctly. Although OpenSARShip is a medium-resolution dataset in which ship samples only possess a few pixels, ITS-ShipFormer still presents superior recognition performance. In terms of recall and precision, the method in this article still achieves an excellent performance. Container ships and tankers have the highest rates of recall, both of which are more than 85%. However, the recall of bulk carriers is only 78.42%, which may be due to their tendency to be mistaken for a container ship. In fact, their class differences are rather small, and both function as ships that carry goods. More refined recognition is a challenging problem.

2) *FUSAR-Ship*: The recognition results of the proposed method under the seven-category FUSAR-Ship are presented in Table III. Due to FUSAR-Ship having a better resolution with more ship detail representations that can be learned, the recognition accuracy of ITS-ShipFormer can reach 90.84%. The recall and precision can reach more than 80% for all seven classes of ships. It is obvious that the confusion of class predictions mainly occurs among fishing, other, and other cargo. It can be explained as the three ship classes have similar geometries. The bulk carrier, tanker, general cargo, and container ship classes reach nearly a state-of-the-art recognition performance, which are 96.15%, 94.63%, 98.84%, and 99.62%, respectively. Through the recognition performance and the analysis aforementioned, the effectiveness of our method is validated.

D. Ablation Study

In this section, extensive experiments are implemented to evaluate the effectiveness of each component in ITS-ShipFormer. For a fair comparison, all subsequent studies are performed with the same settings. As listed in Table IV, the "✓" denotes that the module is utilized, while the "×" means that such a module is not employed. The "×" of MHDLC

TABLE II
CONFUSION MATRIX OF HDSS-NET RECOGNITION RESULTS ON THREE-CATEGORY OPENSARSHIP

Dataset	Predicted		Bulk carrier	Container ship	Tanker	Recall(%)
	True					
OpenSARShip	Bulk carrier		256	63	9	78.42
	Container ship		94	697	17	86.26
	Tanker		10	11	125	85.62
	Precision(%)		71.11	90.40	82.78	Accuracy=84.09%
	F1(%)		74.59	88.28	84.18	

TABLE III
CONFUSION MATRIX OF HDSS-NET RECOGNITION RESULTS ON FUSAR-SHIP

Dataset	Predicted	Bulk carrier	Container ship	Fishing	General cargo	Other	Other cargo	Tanker	Recall(%)
	True								
FUSAR-Ship	Bulk carrier	475	2	4	5	0	5	3	96.15
	Container ship	0	521	1	0	0	0	1	99.62
	Fishing	3	0	378	0	36	54	2	79.92
	General cargo	1	3	0	511	0	0	2	98.84
	Other	0	1	17	0	436	58	8	83.85
	Other cargo	4	2	24	0	56	428	7	82.15
	Tanker	1	0	10	0	6	11	493	94.63
	Precision(%)	98.14	98.49	87.10	99.03	81.65	79.98	95.54	Accuracy=90.84%
	F1(%)	97.14	99.05	83.35	98.94	82.73	79.48	95.08	

TABLE IV
INFLUENCE OF EACH MODULE IN ITS-SHIPFORMER.
BEST RESULTS ARE BOLDED

Modules			Accuracy(%)	
MHDLC	SCSM	Hybrid Loss	OpenSARShip	FUSARShip
×	×	×	77.61	85.79
✓	×	×	78.86	86.80
✓	✓	×	81.98	89.18
×	✓	×	80.89	88.51
×	✓	✓	83.31	90.28
✓	✓	✓	84.09	90.84

means the dynamic local convolution in MHDLC is replaced with standard convolution. The “×” of the discriminative hybrid loss means that only CE loss on the CLS token is utilized. The overall comparisons are listed in Table IV.

It can be seen that different components bring improvement to the network to a certain extent. When only the MHDLC block is utilized, the multihead dynamic convolution can explore local features dynamically to prioritize the discriminative SAR ship features, and the recognition rate can reach 78.86% on OpenSARShip and 86.80% on FUSAR-Ship. With the involvement of SCSM, the recognition accuracies lead further improvement. The recognition performance increases to 81.98% and 89.19% on OpenSARShip and FUSAR-Ship, respectively. This indicates that filtering out the useless, even harmful, sea clutter regions can guide the model’s focus toward the informative ship target areas, which is indispensable in improving the SAR ship recognition performance. When only the SCSM is used in our experiment, the recognition accuracy is 80.89% on OpenSARShip and 88.51% on FUSAR-Ship. Upon further introducing the discriminative hybrid loss, the constraints add on the CLS token, and the

informative tokens can promote better inner-class compactness and interclass separability, boosting the recognition accuracy of OpenSARShip and FUSAR-Ship to 83.31% and 90.28%, respectively. Compared with these models, the full version of our model boosts the recognition performance to 84.09% and 90.84% on OpenSARShip and FUSAR-Ship, respectively, illustrating that the three components help each other to achieve the optimal recognition performance. Through the ablation studies with different configurations, it can be validated that the innovations in our model effectively locate informative tokens and filter out irrelevant sea clutter regions. The innovations also promote the features with increased inner-class compactness and interclass separability to deal with the intraclass diversity and interclass similarity issue.

In addition, to intuitively evaluate our innovations in a qualitative way, t-distribution stochastic neighbor embedding (t-SNE) is employed. In the t-SNE visualization, greater distances between different classes correspond to better feature compactness and interclass separability, indicating higher recognition accuracy. Fig. 5(a)–(d) illustrate the visualization without and with the addition of the innovations on the vanilla model, respectively. It can be observed that there is much confusion in the feature embedding of some classes in Fig. 5(a) and (c). On the contrary, in our complete model alleviate the recognition error. The feature embeddings exhibit increased separation between different classes and enhanced aggregation within the same classes, which are shown in Fig. 5(b) and (d).

E. Visualizations

In this section, we first present the visualization results of the selected informative tokens to demonstrate that our model mainly focuses on ship target regions instead of useless sea clutter regions. Subsequently, heatmaps of the test SAR ship images with and without our proposed method further illustrate

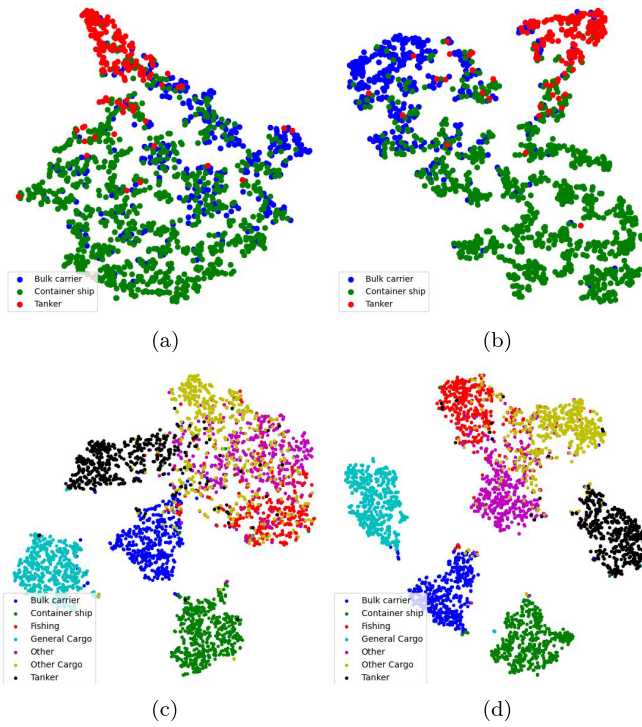


Fig. 5. T-SNE feature distributions of our methods under different configurations on OpenSARShip and FUSAR-Ship datasets. (a) Vanilla method on the OpenSARShip dataset. (b) Our method on the OpenSARShip dataset. (c) Vanilla method on FUSAR-Ship dataset. (d) Our method on the FUSAR-Ship dataset.

that our model can guide the model's attention to the ship target regions and exclude the interference of sea clutter. We conduct the visualization experiments on OpenSARShip and FUSAR-Ship datasets.

1) *Visualizations of the Informative Tokens*: The inherent and unique advantages of the transformer structure lie in its ability to directly capture the importance of all image tokens with respect to class information, thereby selecting and localizing informative tokens for effective and discriminative feature extraction. In ITS-ShipFormer, the SCSM is designed to fully utilize this advantage. To demonstrate our model can effectively focus on informative tokens, we show the visualization results of the informative tokens selected by the third stage and the fourth stage.⁴ The masks consist of the useless tokens determined by SCSM.

- 1) *OpenSARShip*: As shown in Fig. 6, four testing samples from OpenSARShip are selected to demonstrate the effectiveness and robustness of the proposed method. The testing images in the first and second rows do not have sea clutter interference. We can see our model precisely focuses on the informative ship target regions in the image, which is meaningful for making recognition decisions, and drops the uninformative tokens that are useless for recognition. The testing image in the third row has severe speckle noise, and the testing image in the fourth row has strong fog-like radio interference,

⁴In the former two stages of ITS-ShipFormer, the MHDLC blocks are convolution style.

which makes the model hard to pay attention to the ship target regions. We can still find that almost all the informative tokens determined by our model are the most attentive tokens for recognition. By introducing the SCSM, the harmful sea clutter interference regions are removed to guarantee our ITS-ShipFormer concentrates on class-specific tokens in images.

- 2) *FUSAR-Ship*: As shown in Fig. 7, on FUSAR-Ship, we also chose four testing samples to showcase that the proposed method is robust and effective. In contrast to OpenSARShip, the majority of samples in FUSAR-Ship exhibit minimal sea clutter interference; only a few have slight speckle noise interference. Our model demonstrates a precise focus on the informative ship target regions within the image, a crucial aspect for wise recognition decisions, while effectively disregarding uninformative tokens irrelevant to recognition. It can be found that the informative tokens in stage 3 are gathered together, although some of them do not ship target the regions. These informative tokens are composed of cross-shaped artifacts generated by the sidelobes of strong point targets during imaging.

2) *Visualizations of the Heatmaps With and Without Our Proposed Method*: To further illustrate the effectiveness, heatmaps of the test images with and without our method are shown in this section. The samples we utilized in this section are the same as those in Section III-E1. It is clear that the network without our method cannot focus on the informative ship target regions. In some cases, the model without our method even failed to pay attention to the ship target regions and instead focused on the background sea clutter regions, which is useless, or even harmful for recognition. Our method can capture the informative ship target regions and exclude the interference of sea clutter regions to perform more discriminative feature extraction.

- 1) *OpenSARShip*:

From the second line of Fig. 8, although the first and second testing images do not have sea clutter interference, there are also responses on the heatmap in the useless regions. With the proposed method, the corresponding response of the heatmap on the third line only tightly focus on the ship target regions. The third and fourth testing images have strong sea clutter interference, and the response of the heatmap in the second line failed to concentrate on the ship regions. On the contrary, although there are still a few background regions show response, the response of the heatmap with our method can accurately concentrate on the ship target regions.

- 2) *FUSAR-Ship*:

From the second line of Fig. 9, due to FUSAR-Ship having more clear morphological and contour information, the highest response of the heatmap surrounds the ship. Nonetheless, there is still a slight response in the background regions. Contrastingly, in the third row of Fig. 7, the heatmap response generated with our method accurately only focuses on the ship target regions.

TABLE V
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH SOTA METHODS

Model		OpenSARShip				FUSAR-Ship			
		$r(\%)$	$p(\%)$	$f1(\%)$	$Acc(\%)$	$r(\%)$	$p(\%)$	$f1(\%)$	$Acc(\%)$
Transformer-based methods	ViT-B [27]	61.93	67.75	64.71	69.73	74.59	74.23	74.41	74.59
	Swin-S [42]	78.24	70.63	74.24	75.43	84.41	84.47	84.44	84.51
	MobileViT-S [43]	75.66	74.09	74.87	75.98	83.93	84.17	84.05	83.95
	MobileFormer-508M [44]	60.52	65.54	62.93	70.75	84.22	84.56	84.39	84.37
	VAN-B1 [45]	78.33	70.61	74.28	76.44	85.52	85.57	85.55	85.63
	Ceit-S [46]	75.59	73.09	74.31	77.22	85.83	85.91	85.87	85.82
	CvT-13 [47]	64.46	70.88	67.52	73.71	85.00	85.18	85.09	85.09
	Unifformer-S [48]	72.22	75.83	73.98	77.93	86.02	86.09	86.06	86.10
	EfficientViT-M5 [49]	75.88	73.76	74.80	74.34	82.82	83.03	82.92	83.02
	MogaNet [50]	72.79	67.90	70.26	74.34	86.19	86.32	86.25	86.19
	Evit [51]	74.75	69.80	72.19	75.66	84.89	85.08	84.98	84.90
	TransNext [52]	81.55	76.83	79.12	81.36	88.57	88.66	88.62	88.51
Methods for SAR Ship ATR	Wang et al. [53]	57.72	58.72	58.12	69.27	80.09	80.32	80.21	80.02
	Hou et al. [37]	69.93	69.44	66.76	67.41	71.96	72.19	72.08	71.89
	Huang et al. [54]	74.74	69.56	72.04	74.98	83.13	83.34	83.27	83.19
	Zeng et al. [18]	74.99	74.05	74.52	77.41	-	-	-	-
	Xiong et al. [19]	73.87	71.50	72.67	75.44	-	-	-	-
	SE-LPN-DPFF [55]	78.83	76.45	77.62	79.25	-	-	-	-
	MFCFNet [25]	79.37	73.78	76.06	78.60	86.67	86.89	86.69	87.23
	HOG-ShipCLSNet [24]	77.87	72.42	75.04	78.15	86.62	86.54	86.58	86.69
	MFAA [17]	80.38	75.78	78.01	80.27	88.46	88.50	88.48	88.46
	CRID [22]	82.49	79.19	80.81	82.76	89.50	89.62	89.56	89.49
	SA ² Net [16]	76.98	76.90	76.94	80.58	88.24	88.52	88.38	88.28
	HDSS-Net [20]	<u>80.66</u>	<u>81.16</u>	<u>80.91</u>	<u>83.23</u>	<u>90.08</u>	<u>90.15</u>	<u>90.12</u>	<u>90.08</u>
	STFormer [56]	79.50	74.86	77.11	78.55	87.22	87.01	87.12	87.20
	MPSA [57]	84.86	76.76	79.23	82.76	89.65	89.49	89.57	89.60
	MGFE-Net [58]	80.97	76.04	78.43	80.73	88.75	88.61	88.68	88.71
	ITS-ShipFormer(ours)	83.31	81.43	82.36	84.09	90.74	90.99	90.86	90.84

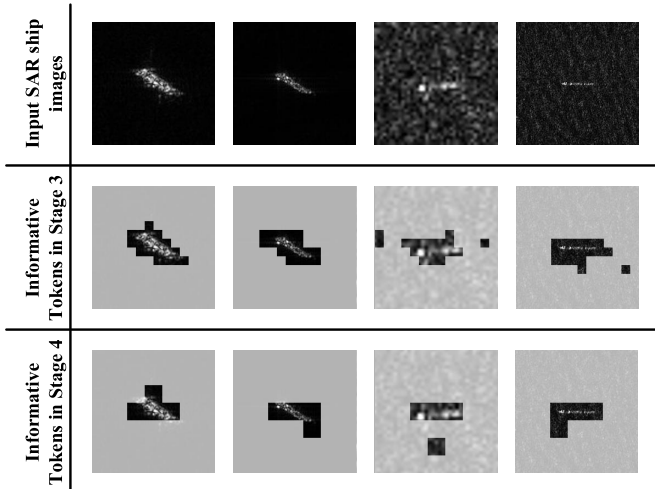


Fig. 6. Visualization results of the informative tokens determined by SCSM on OpenSARShip. The first row shows the original input ship images. The second row shows the informative tokens in the last layer of stage 3. The third row shows the informative tokens in the last layer of stage 4.

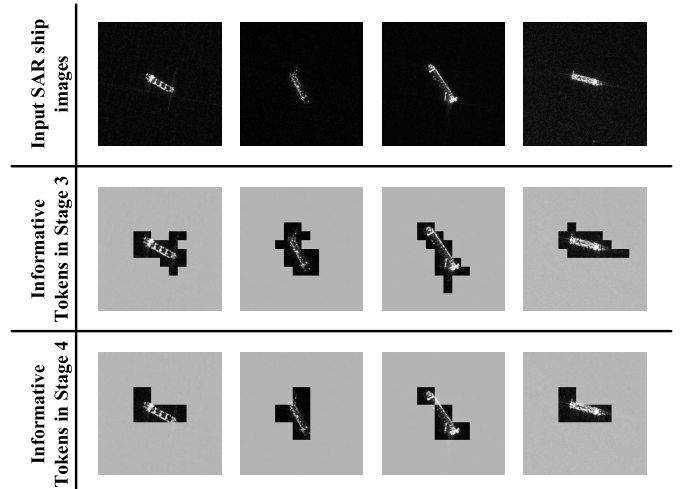


Fig. 7. Visualization results of the informative tokens determined by SCSM on FUSAR-Ship. The first row shows the original input ship images. The second row shows the informative tokens in the last layer of stage 3. The third row shows the informative tokens in the last layer of stage 4.

Nowadays, some works [6], [24], [25], [26], [41] try to incorporate traditional hand-crafted features or physical information to aid the model in making more interpretable

recognition decisions based on SAR images with a clear background. The visualization results demonstrate the effectiveness

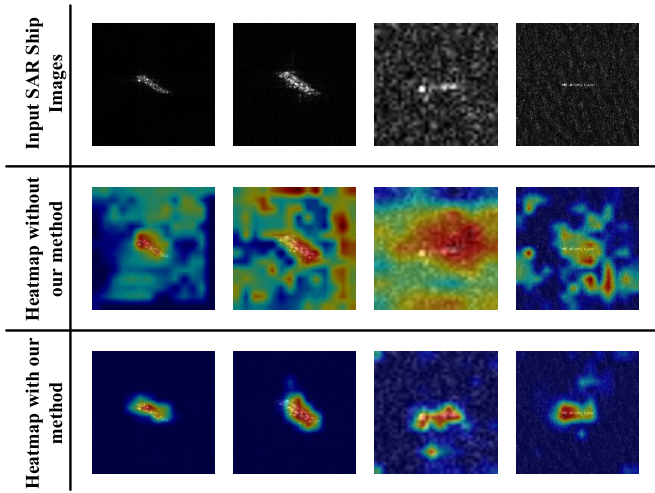


Fig. 8. Visualization heatmap of the testing SAR ship images from OpenSARShip. The first network configuration is the vanilla network without our proposed method. The second network configuration is our method shown in Fig. 3.

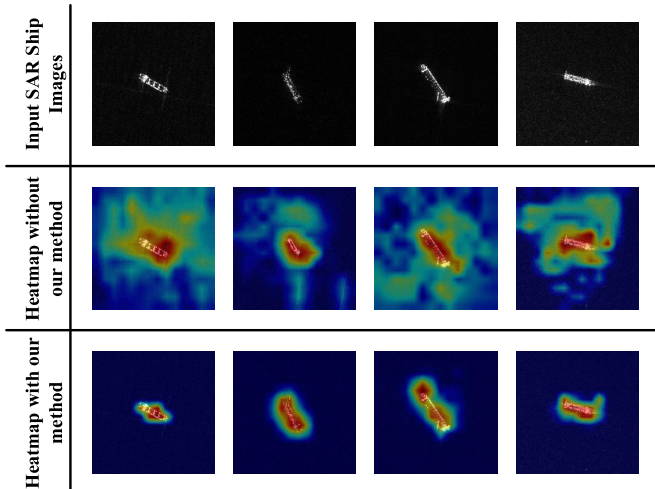


Fig. 9. Visualization heatmap of the SAR ship image from FUSAR-Ship. The first network configuration is the vanilla network without our proposed method. The second network configuration is our method.

of our model and can extend these explorations to a wider range of practical application scenarios.

F. Comparison With SOTA Methods

To comprehensively evaluate the performance improvement of the ITS-ShipFormer, extensive experiments are conducted by comparing the proposed method with CNN-based methods, ViT-based methods, and novel SAR ship recognition methods on OpenSARShip and FUSAR-Ship. Table V lists the comparison of the quantitative evaluation indices to verify the effectiveness of ITS-ShipFormer. We also compare the Params and FLOPs of the proposed ITS-ShipFormer to those of the SOTA methods in Table VI to demonstrate its efficiency. The inputs of Zeng et al. [18], Xiong et al. [19], and SE-LPN-DPFF [55] are paired VV–VH SAR amplitude images. The inputs of others are unpaired VV and VH SAR amplitude

TABLE VI
IMPLEMENTATION COSTS OF DIFFERENT MODELS

Models	Params(M)	FLOPs(G)
Swin-S	48.79	8.54
MobileViT-S	4.99	1.77
MobileFormer-508M	12.06	0.51
VAN-B1	13.90	2.51
Ceit-S	24.20	4.5
CvT-13	19.60	4.07
LiT-S	27.0	4.1
Uniformer-S	22.0	3.6
EfficientViT-M5	12.4	0.52
Wang et al.	15.52	15.35
Hou et al.	2.75	1.97
Huang et al.	5.96	2.68
Zeng et al.	237.03	15.57
Xiong et al.	0.10	0.86
SE-LPN-DPFF	4.22	36.53
HOG-ShipCLSNet	100.23	2.96
SA ² Net	32.45	6.89
HDSS-Net	48.75	5.62
ITS-ShipFormer	9.35	0.907

images, which means single-channel VV and VH SAR images are successively input into networks directly.

From Tables V and VI, the following conclusions can be drawn.

- 1) On the OpenSARShip dataset, ITS-ShipFormer offers the highest recognition accuracy among others. Among all the SOTA methods, the suboptimal recognition is 83.23% from HDSS-Net, which is lower than our ITS-ShipFormer by 0.86%. HDSS-Net ignores the necessity of excluding the interference of useless or even harmless sea clutter regions and guides the model's attention to the informative target regions, blocking to reach higher recognition accuracy.
- 2) On the FUSAR-Ship dataset, ITS-ShipFormer also offers the highest recognition accuracy among others. The suboptimal recognition is 90.08% from HDSS-Net, which is lower than our ITS-ShipFormer by 0.76%. It can be found that although FUSAR-ship has more ship categories than OpenSARShip, the recognition accuracy is higher. This is mainly because FUSAR-Ship has higher image resolution, and more ship detail representations can be learned. We also should note that the FUSAR-Ship dataset only provides single-channel SAR images, so the methods [18], [19], [55], which need dual-polarized SAR images, can not be performed on FUSAR-Ship.
- 3) It should be noted that the proposed model [17] is also trying to capture key image regions that are beneficial for object recognition to improve recognition performance. However, unfortunately, Wang et al. [17] try to capture the key feature regions by training a subnetwork, and there is no loss function to constrain the

discriminative regions. Compared with our ITS-ShipFormer, the captured key regions lack certain interpretability and effectiveness. Therefore, shows a decreasing performance compared to ITS-ShipFormer.

- 4) Nowadays, some explorations [24], [25] try to introduce traditional mature hand-crafted features to assist the model in making better and more explicable ship recognition decisions. However, these explorations need high-quality SAR ship images with no sea clutter interference to extract hand-crafted features. SAR ship images with sea clutter interference undeniably disturb the extraction of hand-crafted features, which is inevitable in practical situations. This is why [24], [25] show worth recognition performance than [16], [20]. The recognition results demonstrate that the proposed method's ability to suppress sea clutter and focus on target regions yields excellent recognition performance. ITS-ShipFormer can lay a solid foundation for the future exploration of fusing information from both handcrafted and deep features to build models.

In summary, the designed ITS-ShipFormer endows the ship recognition framework with enhanced recognition performance and interpretability in an effective and efficient manner. The effective model and hybrid loss function can extract more discriminative features with increased inner-class compactness and interclass separability. The SCSM effectively eliminates sea clutter interference, directing the model's focus toward ship regions. Hence, the proposed ITS-ShipFormer is robust to various conditions, which will boost the practical application of SAR ship ATR methods.

G. More Discussions

1) *Efficiency*: In addition to the accuracy, we also compare the Params and FLOP of the proposed method to those of the SOTA models. From Table VI, we find that the Params and FLOPs of the proposed ITS-ShipFormer are 9.35M and 0.907G, respectively. Compared to the suboptimal HDSS-Net, our model shows accuracy improvement with less than $5\times$ the number of parameters and flops, demonstrating its efficiency and effectiveness.

2) *Future Work*: The detection results of the inshore ship target may not be as expected due to the complex scenarios, such as a few and scattered buildings, containers, and vehicles. To deal with the challenge of the sea clutter issue in SAR ship ATR, our article distinguishes informative tokens and useless tokens. Recently, we have been working to adapt such advantages of the transformer framework for the SAR ship detection task to filter the land regions and inspire the model to mine more discriminative target-related features, leading to more exciting SAR ship detection performance.

3) *Potential Use Cases in Real-World Applications*: ITS-ShipFormer's ability to suppress sea clutter and focus on ship target regions enhances its robustness in challenging maritime environments, such as those with strong speckle noise, ship wakes, or fog-like interference. This makes it highly suitable for applications like maritime domain awareness, illegal fishing detection, and naval defense operations. For instance,

its high accuracy on the FUSAR-Ship dataset and resource-related advantages demonstrate its potential for high-resolution SAR systems like GF-3, which are used in real-time maritime monitoring. In addition, the model's interpretability, as shown through heatmap visualizations, supports decision-making by human operators in mission-critical scenarios.

IV. CONCLUSION

The sea clutter interference and intraclass diversity interclass similarity in SAR ship images present significant challenges for SAR ship ATR, especially the former challenge, which has not been explored yet. A task-specific ITS-ShipFormer is proposed to tackle these challenges. The experimental results and visualization results on benchmark datasets OpenSARShip and FUSAR-Ship show the effectiveness and efficiency of ITS-ShipFormer. From the ablation study, the MHDLC block, the SCSM, and the discriminative hybrid loss have different effects on SAR ship ATR. The MHDLC block replaces MSA with multihead dynamic convolution, which enables the model to enhance the feature extraction ability to deal with the intraclass diversity and interclass similarity. The SCSM guides the model's attention to the SAR ship regions and uses a double flow token updating strategy to concentrate the computations on informative tokens, addressing the sea clutter interference issue in an effective and efficient way. The discriminative hybrid loss adds constraints on both the CLS token and the informative tokens, forcing the model to extract features with increased inner-class compactness and interclass separability. The three innovations of ITS-ShipFormer help each other to achieve optimal recognition performance by dealing with the challenges in SAR ship ATR. However, beyond the two public datasets, our data is not that rich. In the future, further experiments with more diverse samples are needed. We hope the promising results achieved by ITS-ShipFormer can provide a new perspective to rediscover the powerful strength of transformer structure, and our ITS-ShipFormer could be a starting point for future works.

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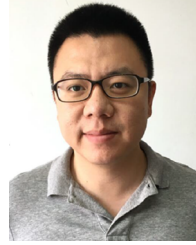
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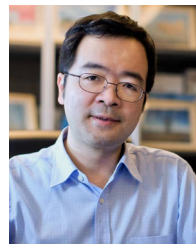
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